CHAPTER

12

Ordinal Instinct: A Neurocognitive Perspective and Methodological Issues

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OUTLINE

As early as 450 BCE, the Greek philosopher Anaxagoras of Clazomenae argued for the existence of an ordering principle in the universe and introduced the concept of Mind as an ordering tool (Burnet, 1908). Interestingly, contemporary scientific and clinical approaches have not availed themselves of this insight. Here, I propose an innovative outlook that argues for the existence of a neurocognitive system designed to evaluate ordinal relationships. This neurocognitive system, termed here "the

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ordinal instinct," is tuned to optimally pick up ordinal information by *implicitly* and *unconsciously* capturing sequential regularities in the surrounding world. Importantly, I review data that may suggest that human numerical intelligence significantly relies on the ordinal instinct through biological development and experience.

12.1 SCIENTIFIC KNOWLEDGE AND DEVELOPMENTS

Numerical cognition is important because mathematical skills are essential for productive functioning in our progressively complex, technological society. In addition, numerical development has been a focus of the continuing theoretical debate concerning the origins of cognition and how it develops throughout life. Approximately 10% of the population has low numeracy skills, a condition called developmental dyscalculia (DD) or mathematical learning disability (MLD) (Kaufmann et al., 2013; Rubinsten & Henik, 2009; Szűcs & Goswami, 2013). Numerical difficulties result in reduced educational and employment achievements, as well as increased costs to physical and mental health (eg, Parsons & Bynner, 2005; Reyna, Nelson, Han, & Dieckmann, 2009). Some argue that in Western societies, poor numeracy is a more significant handicap than poor literacy (eg, Estrada, Martin-Hryniewicz, Peek, Collins, & Byrd, 2004; Parsons & Bynner, 2005). Hence, mathematical skills may have an impact on social mobility and poverty levels. Despite its manifest importance, paradoxically, and compared to other cognitive abilities such as reading and attention, the neurocognitive development of numerical abilities has been neglected by both clinicians and researchers.

12.2 NEURAL AND COGNITIVE FOUNDATIONS OF NUMERICAL KNOWLEDGE

Kant argued that human beings are endowed with innate knowledge that is not related to experience, as part of their rational nature (Kant, 1908). Following such ideas, modern research has argued that mathematics rests on mental representations that developed in the course of evolution (Cantlon, Platt, & Brannon, 2009; Feigenson, Dehaene, & Spelke, 2004). These core representations include a nonsymbolic numerical system that represents the approximate numerical value of a collection of objects (ie, *numerosity*). Nonsymbolic approximation is a key evolutionary ability, which is believed to develop without formal teaching. Indeed, studies have found that infants from birth (eg, Izard, Sann, Spelke, & Streri, 2009) and even nonhuman animals (eg, fish—Piffer, Petrazzini, & Agrillo, 2013) display the ability to discriminate quantities. In contrast, symbolic and exact numerical representations (eg, Arabic numerals such as 6) vary across cultures, are influenced by language, and arise late in human development (as opposed to nonsymbolic approximates).

Both exact symbolic and approximated nonsymbolic numerical knowledge is most commonly measured by comparison tasks, in which participants are asked to identify the larger among two arrays of dots. One major signature of nonsymbolic core numerical representations that is present in human and nonhuman animals is that comparisons are subject to a ratio (minimum/maximum) limit: Accuracy decreases while reaction time (RT) increases as the ratio of the compared numbers approaches one (ie, the ratio effect; Cantlon et al., 2009). This ratio effect is the result of both the distance between two quantities and their location on the mental number line. Similarly, the larger the distance between two numbers to be compared the faster the response (ie, the distance effect; eg, Ansari, 2008). The distance effect was first reported by Moyer and Landauer (1967) and has since been replicated by many researchers under a variety of conditions (eg, Dehaene, 1989; Kucian, Loenneker, Martin, & von Aster, 2011). The numerical distance effect has been found in children (eg, Holloway & Ansari, 2009; Landerl & Kölle, 2009; Sekular & Mierkiewicz, 1997) and in primates (Nieder, Freedman, & Miller, 2002). Similarly, the ratio effect has been demonstrated in infants (Xu & Spelke, 2000; Xu, Spelke, & Goddard, 2005), in young children (Barth, La Mont, Lipton, & Spelke, 2005), and in animals (eg, Cantlon & Brannon, 2007; Hauser, Tsao, Garcia, & Spelke, 2003). Both of these signature effects (ratio and distance) are compromised in developmental dyscalculia (DD) (for the distance effect, see Heine et al., 2013; Mussolin et al., 2010; Price, Holloway, Vesterinen, Rasanen, & Ansari, 2007; Soltesz, Szucs, Dekany, Markus, & Csepe, 2007) (for the ratio effect, see Kovas et al., 2009; Libertus, Feigenson, & Halberda, 2011). Further research identified the neural tissues involved in ratio and distance effects. Specifically, the parietal lobes and in particular the intraparietal sulcus (IPS) serve the mental operations involved in these effects (eg, Castelli, Glaser, & Butterworth, 2006; Cohen Kadosh, Bien, & Sack, 2012; Cohen Kadosh, Cohen Kadosh, Kaas, Henik, & Goebel, 2007; Fias, Lammertyn, & Reynvoet, 2003; Piazza, Izard, Pinel, Le Bihan, & Dehaene, 2004).

This accumulated body of results has led to an accepted view of the approximate numerical system (numerosity) acting as an innate foundation of numerical knowledge (eg, Butterworth, 2010). Moreover, it has been suggested that DD involves a deficit in this foundation of knowledge (eg, Butterworth, 2010) (but see Kaufmann et al., 2013; Rubinsten & Henik, 2009).

However, several findings and methodological issues suggest that this wide agreement should be reexamined. (1) A major obstacle to the study of cognitive and neural correlates of numerical knowledge is encompassed in the difficulty of teasing apart, at the cognitive level of analysis, processes which are involved—to varying degrees—in both ordinal and quantity processes. Under normal conditions, order and quantity are inseparably bound. To be able to estimate which of two quantities is larger, one needs to know the position of the specific quantity in relation to other quantities, that is, ordinal information. Specifically, the ratio and distance effects, which are considered signatures of a phylogenetically ancient core quantitative system, are a result of the ability to assess ordinal relationships and not only quantities.

(2) Occasional reports show that the ability to process ordinality was found in primates (eg, Brannon, Cantlon, & Terrace, 2006; Cantlon & Brannon, 2006) and in other animal species phylogenetically distant from humans, such as rats (Suzuki & Kobayashi, 2000) and parrots (Pepperberg, 2006). Infants' ability to detect ordinality was found with size-based sequences (De Hevia & Spelke, 2010; Macchi Cassia, Picozzi, Girelli, & de Hevia, 2012), arbitrary sequences of different items (eg, Lewkowicz & Berent, 2009), and with quantities (eg, Brannon, 2002; De Hevia & Spelke, 2010; Feigenson, Carey, & Hauser, 2002; VanMarle, 2013) (Gevers, Reynvoet, & Fias, 2003). These findings provide clues to the existence of an evolutionary-based ordinal input analyzer. However, these findings are typically intepreted such that participants are able to actively compute ordinal information or to actively learn how to place stimuli based on their relative ordinal position and that this skill plays a role in numerical cognition. For example, it has been shown that nonhuman primates can learn to use ordering rules similar to those that emerge during human development (McGonigle & Chalmers, 2006).

In contrast to the argument that ordinal perception is learned during development, an alternative interpretation to previous findings and arguments may be suggested. That is, it may be argued that humans have a neurocognitive platform of order, which is termed here the "ordinal instinct." This ordinal instinct is at the heart of the preverbal ability to *implicitly*, *inattentively*, and *immediately* perceive ordinal relationships (as opposed to active, attentive, and exact computation) between any types of stimuli in the visual scene. In current opinion, numerosity (not order) is agreed to be at the heart of the numerical core system. In addition, previous research claimed that ordinality is at work in solving unique numerical problems such as counting (Gallistel & Gelman, 1992), but, to the best of our knowledge, the ability to process ordinal information is not currently recognized as a system that evolved under selection pressures for ordinal representations. Thus, the fact that at later stages of development ordinality served other purposes, including numerical representations, is not acknowledged by contemporary science. Despite the fact that much of our perceptual abilities, which are highly important for survival, these perceptual abilities are all based on lower-order geometric properties, such as ordinal relations. Examples include perceiving objects in three-dimensional space (Koenderink, van Doorn, Kappers, & Todd, 2002; Norman & Todd, 1998), face recognition (Gilad, Meng, & Sinha, 2009), and movement perception. That is, even highly complex tasks such as tossing an object to a target, could be performed simply based on an ordinal representation of depth and a corresponding ordinal representation of limb force (Todd & Norman, 2003). Moreover, possible neural constructs have been suggested for such an ordinal mechanism. Neurons in the early stages of the mammalian visual pathway provide a plausible substrate for extracting ordinal image relationships (Albrecht & Hamilton, 1982). In addition, Sawamura, Shima, and Tanji (2002) found single-cell neurons located in the superior lobule in the parietal cortex of monkeys that are selective for ordinal position in sequences of arm movements. Hence, it may be suggested that the evolution and development of brain systems critically involved in the perception of action, space, and numerical cognition (Walsh, 2003) depend on an initial ability to evaluate order. That is, it may be speculated and suggested as a theoretical framework that there are two separate coexisting systems: a developmentally older ordinal system and a separate numerosity system. It may be proposed that the neurocognitive system that manages the ordinal instinct might have had a major role in the development of the occipitoparietal dorsal brain system, known to be involved in the perception of action, spatial locations, and numbers. As mentioned earlier in this chapter, ordinal information is indeed highly linked with perceiving action (eg, Sawamura et al., 2002), space (eg, Gilad et al., 2009), and numbers. Accordingly, the ordinal instinct might be also involved with setting up neural and cognitive networks of the numerosity system.

Indeed several recent findings point to specific brain areas that are involved with abstract representation of ordinal information that is not necessarily number specific. Two recent papers (Fias, Lammertyn, Caessens, & Orban, 2007; Ischebeck et al., 2008) compared the neural bases of symbolic ordinality and numerical processing by using functional magnetic resonance imaging (fMRI). Fias et al. (2007) contrasted brain activation during symbolic numerical comparisons (eg, which comes first—3 or 9) with brain activation during comparisons of nonnumerical stimuli that carry ordinal information. Ischebeck et al. (2008) compared word generation of numbers, months, both denotes ordinal information, with word generation of animals which do not have ordinal meaning. These studies found that the anterior intraparietal sulcus [IPS; specifically, the horizontal segment of the IPS (hIPS) in Fias's study and in Ischebeck study, it was the hIPS2 which is the most anterior and lateral part of the IPS] responds equally to both numerical and nonnumerical ordinal information. Similarly, Kaufmann, Vogel, Starke, Kremser, and Schocke (2009) found that in children the processing of numerical (symbolic) and nonnumerical (sizes) ordinality is supported by the IPS. Collectively, these findings suggest that the IPS and, specifically, the anterior region of the IPS may be involved in the abstract representation of ordinal information that is not number specific. Hence, there are domain-general representations of ordinal information that are involved with any type of stimulus that embodies ordinal information such as numerical, magnitude, and alphabetical stimuli. However, Zorzi, Di Bono, and Fias (2011) used support vector machines to reanalyze the data of Fias et al. (2007), who found a clear dissociation between processing numerical versus alphabetical orders in bilateral horizontal IPS. These findings support previous neuropsychological studies with brain-damaged patients (eg, Delazer & Butterworth, 1997; Turconi & Seron, 2002), and suggest, in contrast to the Fias or Ischebeck arguments (Fias et al., 2007; Ischebeck et al., 2008) that ordinal and quantity processing dissociate at both the behavioral and biological levels.

Accordingly, scientific evidence is inconclusive; some evidence suggests that a single numerical magnitude system operates over both quantity and ordinality information, whereas other evidence shows signatures of separate cognitive systems of ordinality and quantity. One reason for the lack of clarity in previous results (eg, Fias et al., 2007 vs. Ischebeck et al., 2008) could be the use of different methodologies in the study of ordinality. I turn to further discuss this issue.

12.3 METHODOLOGICAL ISSUES IN STUDYING ORDINALITY

Ordinality is considered here as a segment of the visual scene at a given location that is perceived by an observer to have a beginning and an end of sequential relationships between countable or noncountable items. I call the process by which observers identify these beginnings and endings and their relationships—order perception, ordinality, or *the ordinal instinct*. This concept suggests that although ordinality is in the mind of the beholder, it is tied to visual scenes (Zacks & Tversky, 2001). Hence, effects of both environment, biology, and cognition should be investigated.

Beside quantities, digit symbols have their own ordinal features that may modulate their effect (eg, when they produce a conflict and are presented in a nonordered fashion). These ordinal features could be employed to ask questions about digit or quantitative representations.

We (eg, Rubinsten & Sury, 2011; see Fig. 12.1) and others (eg, Brannon & Terrace, 1998; Franklin, Jonides, & Smith, 2009; Kaufmann et al., 2009; Lyons & Beilock, 2011, 2013; Macchi Cassia et al., 2012; Merritt, MacLean, Jaffe, & Brannon, 2007; Terrace, Chen, & Jaswal, 1996) have been using similar situations to study ordinality. The basic paradigm presents participants with *n* (more than 2) stimuli (eg, groups of dots—Fig. 12.2; or Arabic numerals, shades of gray, size, etc.) in an ordered or nonordered form (eg, groups of dots—see Fig. 12.1; Arabic numeral, shades of gray, different

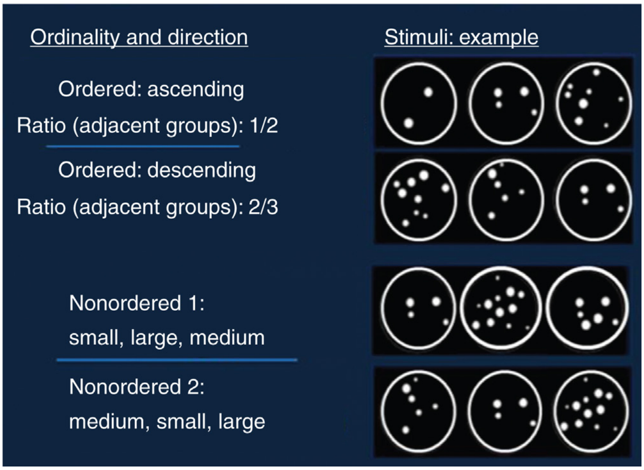

FIGURE 12.1 Examples of stimuli (each stimulus is represented by one *black rectangle* that includes three simultaneously presented *circles with dots*). The task is to decide whether the three circles include dots that are presented in an ordered or nonordered fashion (in the current example, based on quantity).

physical sizes of arbitrary stimuli, etc.). Participants' task is to indicate whether the stimuli are ordered or not.

Counting is not possible due to the brief presentation. Specifically, the ordinal instinct should be studied similar to investigations of numerical estimations. That is, estimation of numbers relates to the strategy employed when a stimulus configuration is comprised of a large number of items and is presented briefly (Pavese & Umiltà, 1998). It is an intuition

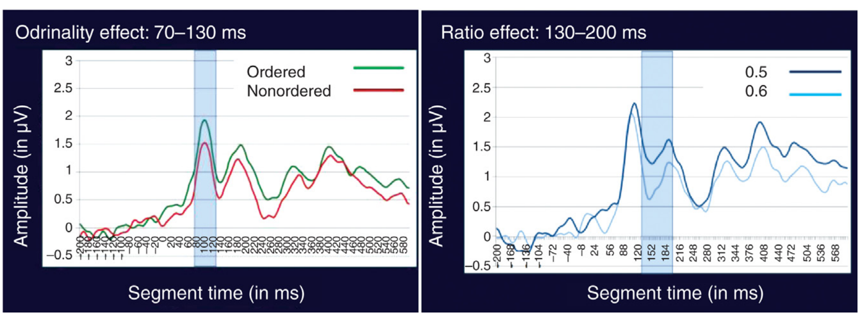

FIGURE 12.2 An ERP study showed that the P1 component, between 80 and 130 ms after stimulus onset (left), was modulated by ordinal information regardless of ratio. A later positive component (right), separated from the ordinal effect, ranging roughly from 130 and 200 ms, was sensitive to the "ratio" manipulation (larger amplitudes in response to smaller ratios as compared to larger ratios).

available to humans regardless of language and education; hence, numerical estimation is considered to be part of the core numerical system (Dehaene, 2009) that is innately available to humans and nonhuman animals (Cantlon et al., 2009). Therefore, to study the ordinal instinct as a core ordinal system, critical variables should include nonsymbolic and approximations of ordinality (as opposed to linguistic/symbolic counting).

In addition, in most of the studies in the field of ordinal processing, participants are presented with pairs of items (eg, numbers, letters, months, etc.) and are asked to decide whether these pairs are presented in an ascending or descending order (eg, Fias et al., 2007; Turconi, Campbell, & Seron, 2006) or to decide which one of the items appears before/after in a sequence (eg, Brannon & Terrace, 1998; Brannon & Van de Walle, 2001). In contrast, presenting three or more items as a single stimulus, and forcing participants to pay attention to all the numbers or quantities as a triad or more (eg, Franklin et al., 2009; Kaufmann et al., 2009; Rubinsten, Dana, Lavro, & Berger, 2013; Rubinsten & Sury, 2011) can better emphasize the ordinal relationship of a sequence. The study of ordinality can be compared to statistical methods for measuring and analyzing change (eg, change such as ascending, descending, ordered, nonordered). Such statistical methods for change include individual growth modeling (eg, Rogosa & Zimowski, 1982), hierarchical linear modeling (Bryk & Raudenbush, 1992), random coefficient regression (Hedeker, 2015), and multilevel modeling (Goldstein, 2011). These statistical methods require at least three points of measurement. This suggests that change (including order perception) cannot be viewed as the difference between "before" and "after" or "smaller" and "larger." That is, two points, which are typically used by modern science in investigations of ordinality, may not be sufficient to reveal the features of the trajectory. Hence, to better separate ordinal from quantity processing, the basic paradigm should present participants with *n* (but more than 2) stimuli (eg, groups of dots—Fig. 12.1) that are displayed briefly and simultaneously.

Why simultaneously? Order and timing are dissociate (as was shown, eg, by Schubotz & von Cramon, 2001). For example, it had been shown that timing and ordering are closely related but act as independent processes in the programming of movement (eg, MacKay, 1985). Also, when presenting stimuli as a dynamic sequence along a timeline, additional cognitive processes are being imposed by the task structure, such as short-term memory (or working memory). Short-term memory is capacity limited, and encoding into explicit short-term memory requires attention (Rensink, 2000; Sperling, 1960). Hence, in the investigation of ordinality, when presenting variables in a dynamic presentation, there is a need to actively initiate order encoding (Holcombe, Treisman, & Kanwisher, 2001; Potter, 1976) (as opposed to implicit nonattentive approximation in ordinal instinct). Indeed, in the numerical cognition field, it had been shown that when the elements of a set are presented one after the other, they are enumerated consecutively (Cordes, Gelman, Gallistel, & Whalen, 2001), a process that requires symbolic counting, which is based on learned number symbols. In contrast, when presented simultaneously as in multiple-item patterns, numerosity can be estimated at a single glance (eg, Barth, Kanwisher, & Spelke, 2003) and show equal numbers of scanning eye movements (Nieder & Miller, 2004).

Examples for studies that used such a methodological paradigm have been given by Brannon and Terrace (1998) who presented four items of different sizes and shapes to monkeys; Terrace et al. (1996), who presented three-colored fields of light, achromatic geometric shapes, and lines' orientation to pigeons; Merritt et al. (2007) who used three and four digitized color photographs to *Lemur catta*; Kaufmann et al. (2009) who presented three Arabic numerals or symbols varying in font sizes to children with dyscalculia; Lyons & Beilock (2013) who presented three groups of dots, Arabic numerals, and luminance to intact developing university students and three Arabic numerals in university students (Lyons & Beilock, 2011); Macchi Cassia et al., 2012, who presented three sizes to 4-month-old infants; also by Rubinsten and Sury who presented three groups of dots to university students with dyscalculia (Rubinsten & Sury, 2011) and Franklin et al. (2009), who used three double-digit numbers or names of months.

In addition, it should be noted that controlling or manipulating multiple visual parameters is not an easy task since there is a considerable correlation between numerosity and many other different visual cues (eg, Gebuis & Reynvoet, 2012; Leibovich & Henik, 2013). Accordingly, density and aggregate surface (area) are typically being controlled within the previously mentioned methodology, but this aspect (ie, visual cues) should be further investigated. Specifically, multiple visual parameters should be better controlled (ie, by being kept constant or randomized) or manipulate simultaneously. This may allow investigators to study whether participants process ordinal information of the visual cues even though they are uninformative about numerosity. In such case, it may be concluded that ordinality acts, as an instinct that enables implicit analysis of the visual scene.

Finally, and as can be seen in Fig. 12.1, to dissociate ordinality from numerosity, we systematically manipulate the ratio between adjacent stimuli. We have shown that such a methodology (eg, Fig. 12.1) is capable of teasing apart numerosity and ordinality, at both the cognitive and biological levels (Rubinsten et al., 2013). That is, in an event-related potentials (ERP) experiment, we (Rubinsten et al., 2013) found that regardless of the ratio between the three groups of dots, the amplitude of the ERP wave, between 80 and 130 ms after stimulus onset (known as the P1 component), was modulated by the ordinal information (ie, ordered vs. nonordered stimuli) and showed a restricted posterior occipitoparietal distribution. Ratio effect, on the other hand, began and ended later. That is, a later positive component, separated from the ordinal effect, ranging roughly from 130 to 200 ms, was sensitive to the "ratio" manipulation: it showed larger amplitudes in response to smaller ratio between the three circles, compare to larger ratios.

To note, the P1 component has been associated primarily as an early sensory/perceptual responses. It also should be noted that the current ordinal effect has been found to resist extensive manipulation of the nonnumerical parameters of the display, thus evading simple explanations in terms of density, or area. That is, no visual analysis (eg, density or area) was possible other than ordinal information. Accordingly, and taking together the behavioral as well as the ERP data, it was argued that P1 component is also associated with an early visual mechanism dedicated to analyzing ordinal information and providing a sensory representation of order to higher-level perceptual systems (eg, systems that process quantity or direction). Hence, it may be suggested that ordinal relationships are perceived earlier than numerosity, independent of the ratio between quantities. In addition, ordinality is processed very early after stimulus presentation, too early to be attentively or explicitly processed, supporting the claim that ordinality is implicitly perceived (Fig. 12.2).

Yet, to best capture the neural and cognitive signature of the ordinal instinct, within the basic paradigm described earlier (Fig. 12.1), the issue of nonsymbolic versus symbolic presentations should also be taken into account. I elaborate later.

12.4 SYMBOLIC VERSUS NONSYMBOLIC REPRESENTATION

Symbolic numerical representations are distinct, accurate, verbally based, culturally dependent, and require explicit learning (eg, Arabic numerals such as "6" or number words such as "SIX").

Specifically, very recent evidence, show that active symbolic number-ordering acts as an important factor in the aquisition of symbolic numerals (Lyons & Beilock, 2009). For example, children's performance in a symbolic ordering task, in which they are asked to decide if three written Arabic numerals are in increasing order or not, better predicts their symbolic arithmetic performance in grades 1 to 6, than number comparison (Lyons, Price, Vaessen, Blomert, & Ansari, 2014). Such findings support the role of active symbolic ordering in the acquisition of symbolic numerals.

In contrast, the ordinal instinct targets the nonsymbolic, preverbal, and approximate aspect of ordinality (ordinal instinct) as the basis of higher-order mathematics. Even Charles Ransom Gallistel and Rochel Gelman, the influential pioneers in the field of numerical neurocognition, argued that "The preverbal process for comparing magnitudes (preverbal ordination) renders the ordering of the verbal numbers intelligible" (Gallistel & Gelman, 1992).

It is clear that education enables the shift from an informal approximate representations of ordinality to a formal, symbolic, exact representation of ordinal relationships (eg, of Arabic numerals in the counting system; eg, Lyons & Beilock, 2009). Lyons and Beilock (2009) showed that in learning settings, working memory capacity is highly important for linking the ordinal representations of symbolic numerical with nonsymbolic numerical information. Specifically, in their first experiment, Lyons and Beilock (2009) trained adult participants to associate dot quantities with novel symbols and tested, among other things, the ordinal perception of the novel symbols. Findings showed that participants who were higher in working memory capacity learned ordinal information about the symbols that lower working memory individuals did not. In addition, different languages, which are symbolic by definition, have an important role. Accordingly, the direction of the writing system, for example, may have a significant impact on the processing of ordinal information. Indeed, we (eg, Hochman Cohen, Berger, Rubinsten, & Henik, 2014; Prior, Katz, Mahajna, & Rubinsten, 2015), have shown significant differences between Hebrew and Arabic native speakers in processing numerical information (not ordinality) due to the different directions of the numerical writing systems (ie, in contrast to Arabic numerals, Indian numerals, used by many Arabic-speaking peoples, are written from right to left). In addition, we have reported (Sury, Rubinsten, De Visscher, & Noël, submitted) a cross-linguistic experiment of ordinal processing exploring the relationship between ordinal and numerical information. Two groups of Hebrew-speaking Israeli and French-speaking Belgian children (6–9-year-olds) completed ordinal judging tasks with quantities (ie, groups of dots) while the direction of the sequence (ascending/descending) was manipulated. Findings indicate that both groups estimate order from 6 years of age (as indicated by the ordinal effect, which is measured by differences in performances to ordered vs. nonordered stimuli). However, responses to ordinal sequences were modulated by direction; Hebrew-speaking children processed ordinal descending direction more accurately (compatible with the right-to-left Hebrew-writing system). In contrast, French-speaking children were more accurate in processing the ascending sequences. That is, the ability to estimate order (ie, the ordinal instinct) exists from an early age in both languages, but linguistic information (eg, direction of the writing system) modulates the development of the ability to estimate ordinal information.

To note, the aforementioned findings, about differences and similarities between ordinal and numerical perception, are all related to intact developing populations. Accordingly and as will be discussed in what follows, it may be very interesting to ask questions about developmental patterns of ordinality and numerosity of cases with numerical deficiencies (DD) and math anxiety.

12.5 ORDINAL INSTINCT AND DEVELOPMENTAL DYSCALCULIA

There are several different lines of evidence for the biological and cognitive precursors of DD. Specifically, specialized neural circuits for numerical processing have been identified in the parietal lobes of the brain, especially the left and right intraparietal sulci (Cohen Kadosh et al., 2007b) and the left angular gyrus (Ansari, 2008; Fias & Fischer, 2005). People with DD show strikingly poor performance on very simple tasks such as difficulty in retrieving arithmetical facts, using arithmetical procedures (Shalev & Gross-Tsur, 2001), and the use of immature problem-solving strategies, even in adulthood (eg, using finger counting—Jordan, Hanich, & Kaplan, 2003). DD is also reflected in deficiencies in basic numerical functions such as comparing nonsymbolic numerical quantities (eg, Halberda, Ly, Wilmer, Naiman, & Germine, 2012; Piazza et al., 2010), processing numbers symbolically (eg, in Arabic notation—Stock, Desoete, & Roeyers, 2010), recognizing patterns and subitizing (Ashkenazi, Mark-Zigdon, & Henik, 2013) and visuospatial working memory (for review, see our paper Kaufmann et al., 2013; Rotzer et al., 2009). For example, Price et al. (2007) found a weak IPS activation in DD children compared to controls when comparing nonsymbolic numerical stimuli. Moreover, Landerl and Kölle (2009) found that 8–10-year-old DD children were slower than controls in both symbolic and nonsymbolic number comparisons. Mussolin et al. (2010) found that 10- and 11-year-old children with DD showed larger distance effect in both symbolic and nonsymbolic numerical comparisons. Also, using the event-related potentials (ERPs) methodology, Soltesz et al. (2007) found that, compared to controls, adolescents with DD show no late event-related brain potentials (ERPs) distance effect between 400 and 440 ms on right parietal electrodes when comparing Arabic numerals. Such findings suggest that the processing of numerical information and representations of the distances between them is abnormal in DD.

Collectively, the behavioral and electrophysiological evidence points to a deficient mental number line, in which quantities are spatially mapped based on distances, as one of the possible outcomes or sources of DD. However, all of these cognitive skills may be also significantly linked with ordinal processing. Accordingly, and as previously suggested, it is possible that the precursor of DD (Rubinsten & Henik, 2009) is a difficulty in the processing and evaluation of ordinal relationships (the ordinal instinct), which may lead to deficiencies in higher mathematical abilities. Indeed, we (Rubinsten & Sury, 2011) showed that adult DD participants exhibit deficiencies in processing ordinal information. Specifically, DD participants had a normal ratio effect in the nonsymbolic ordinal task (Fig. 12.1), regardless of the perceptual condition (ie, constant area or constant density conditions but also in randomized area and density condition). In addition, in this nonsymbolic task, DD participants did not show the ordinality effect in constant area or in density conditions. However, when both cues (ie, area and density) were not controlled (ie, randomized), DD participants showed the ordinality effect, but only in the descending direction. In the symbolic task, the ordinality effect was modulated by ratio and direction, which are more natural to symbolic representations. These findings suggest DD may be deficient in their ability to process ordinal information but also that there might be two separate cognitive representations of ordinal and quantity information and that linguistic abilities may facilitate estimation of ordinal information.

12.6 CONCLUSIONS

The suggested ordinal instinct may have a number of implications for the study of ordinality and numerical cognition. First, it may be suggested that quantities and ordinal information, at least at a certain stage of cognitive processing, are distinct and are not necessarily "two sides of the same coin" (Jacob & Neider, 2008). Second, if human beings are indeed able to estimate order as part of their core cognitive system, it will mean that among some, such as those with developmental dyscalculia, this ability might be deficient. In such a case, those people are partially "blind" to ordinal information and not only to quantity. This hypothesis needs to be investigated because (1) it will help uncover the neurocognitive mechanisms that are basic to DD, and (2) it will shed light on the important issue of how numerical cognition, and more specifically the ordinal instinct, can be promoted by classroom practice in typically as well as atypically developing children.

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